but can see what happens down the line. Say, when supply of cotton goes down in the rainy season, or when demand spikes during Diwali, you can tweak the system to meet these contingencies and test the solution in the virtual world, before making the change for real.

Metaverse in Industry 4.0

"We think of Industry 4.0 as an endeavour to build a connected enterprise that creates an ecosystem of a smart factory, connected workers, connected customers, connected products, and smart services. In such a connected enterprise ecosystem, there is a natural convergence between the physical and digital world. The metaverse is expected to play an important role in enabling this 'phydigital' convergence. The nature of the metaverse experience for industrial enterprises would vary based on the nature of the company and its products and services. Some common trends and themes that we expect to transform into popular use cases are employee engagement, employee training, product demonstration centres, virtual events, product digital twins, product design collaboration, and multi-party interactions," says Garg.

Immersive human-machine interaction, exemplified via a digital twin is a good reflection of such a paradigm. These digital twins can be of different types: process, product, or system. Each of these types of digital twins can be at varying levels of complexity—from a simple replica of a physical world item to a dynamic bi-directional exchange of data between a physical machine and its digital twin.

"The fourth industrial revolution provides the opportunity to connect the physical and digital worlds aiming to accelerate change in technology, aligning to the increase in interconnectivity and automation across the business ecosystem. The digital twin provides a virtual representation to an actual or physical asset (product and/or system). The advancements in the industrial metaverse take this further, allowing a new representation of products and systems in a new channel of delivery expanding the market and market awareness to a new level. This may also provide prospects with the ability to virtually see and use the product or offering prior to purchasing," says Paul Nashawaty, senior lead analyst, Application and Infrastructure Modernization, Enterprise Strategy Group, a division of TechTarget.


Coca Cola Hellenic Bottling Factory uses smart glasses for vision picking, remote support, and digital training (Courtesy: TeamViewer)

Who should go for it and why?

Actually, everyone—organisations small and large—can think of implementing the industrial metaverse. It is not doable overnight; the industrial metaverse is a long-term pursuit, but it is worth it because the benefits are also longstanding. Thankfully, this is not a tech implementation that you have to finish in one shot—the train can come later, but you can start laying the track today.

Hendrik Witt, chief product officer, TeamViewer, says, "Industrial metaverse technology can be harnessed and applied to any industry and across the entire industrial value chain to enable hands-free work with digital information in order to gain speed, improve efficiency and productivity as well as lower costs and error rates. Be it in onboarding and training, logistics, assembly, maintenance, or after-sales service."

"Some of the initial benefits we see of adopting the industrial metaverse are in design and engineering, testing and validation, and training. In design and engineering, the industrial metaverse can enable the entire organisation to get immediate insight in the impact of design changes by distributed teams on all different disciplines. For testing and validation, the combination of photo-realistic environments with multi-physics simulations enables to test all kinds of scenarios and training autonomous systems through machine learning and synthetic data. And when it comes to training and closing the skills gap, the industrial metaverse will enable virtual training for all kinds of scenarios or hazardous situations and enable remote access to expert skills, empowering non-expert employees to solve problems," says Dale Tutt, vice president of Industry Strategy, Siemens Digital Industries Software.

"Some early adopters in the industry are exploring its potential to uncover new interaction and business models. For instance, a multinational oil and gas company is training its employees for safety in deep water oil production with virtual reality (VR). The company is using VR to transfer knowledge from one team to another and saving enormously on time. A leading defence equipment manufacturer is using smart glasses for 30% faster outcomes and 93% enhanced accuracy. In pharmaceutical warehouses, professionals are now assisted by head-mounted gear that mitigates safety risks, allows remote support, and aids in quality assurance by facilitating remote audits and process improvement. Automobile heavyweights are using metaverse precursor technologies to create and test designs collaboratively," says Immanuel J. Kingsley, metaverse technology officer, and head of Innovation Lab at Hexaware Technologies.